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Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I'll start by examining the dataset to understand its structure and key characteristics. Then, you can provide a specific analytical question for me to address.

Let me first load and inspect the data.



Dataset Overview

- **Size:** 231,636 rows × 19 columns
- **Columns:** The dataset includes financial data on municipal contracts, such as funding, investment, commercialization, and industrialization values.
- **Key Numeric Fields:**
 - `FundingValue`, `InvestmentValue`, `CommercializationValue`, `IndustrializationValue`, and `ContractTotalValue` capture different aspects of financial transactions.
 - `MonthIssued` indicates when the contract was issued.
- **Categorical Identifiers:**
 - `StateCode`, `MunicipalityCode`, `ProgramCode`, `EconomicActivity`, and `FinancialInstitutionCNPJ` categorize the contracts by location, program, and financial institution.



What specific analytical question would you like me to address using this dataset? [?]

How elastic is the total contract value in response to changes in the investment area?

To measure the elasticity of **Contract Total Value** in response to changes in **Allocated Area Investment**, I will compute the **elasticity coefficient**, defined as:

$$E = \frac{\% \Delta \text{ContractTotalValue}}{\% \Delta \text{AllocatedAreaInvestment}}$$



This will tell us how sensitive contract values are to changes in allocated investment areas. I'll proceed with calculating this coefficient.

It looks like the elasticity calculation resulted in `NaN`. This might be due to the data containing zero or constant values in `AllocatedAreaInvestment`, which prevents meaningful percentage change calculations.

I'll investigate further to determine if there's enough variability in the data for a valid elasticity analysis. [?]

Findings:

- **Allocated Area Investment Distribution:**
 - **Mean:** 14.5
 - **Standard Deviation:** 142.26 (high variability)
 - **Minimum:** 0.0 (many zero values)
 - **Median (50th percentile):** 0.0
- **Unique Values:** 13,136 distinct values in 231,636 rows (suggests some variation).



Issue Identified:

A large portion of `AllocatedAreaInvestment` values are **zero**, meaning most contracts do not allocate area for investment. Since percentage change calculations require positive values, the dataset lacks enough valid

data points for elasticity estimation.

Next Steps:

Would you like me to proceed by:

1. **Fitting a regression model** to estimate the relationship between `AllocatedAreaInvestment` and `ContractTotalValue` ?
2. **Filtering out near-zero values** and attempting elasticity calculation with only significant investment allocations? [\[>\]](#)

Message Data Analyst



ChatGPT can make mistakes. Check important info.